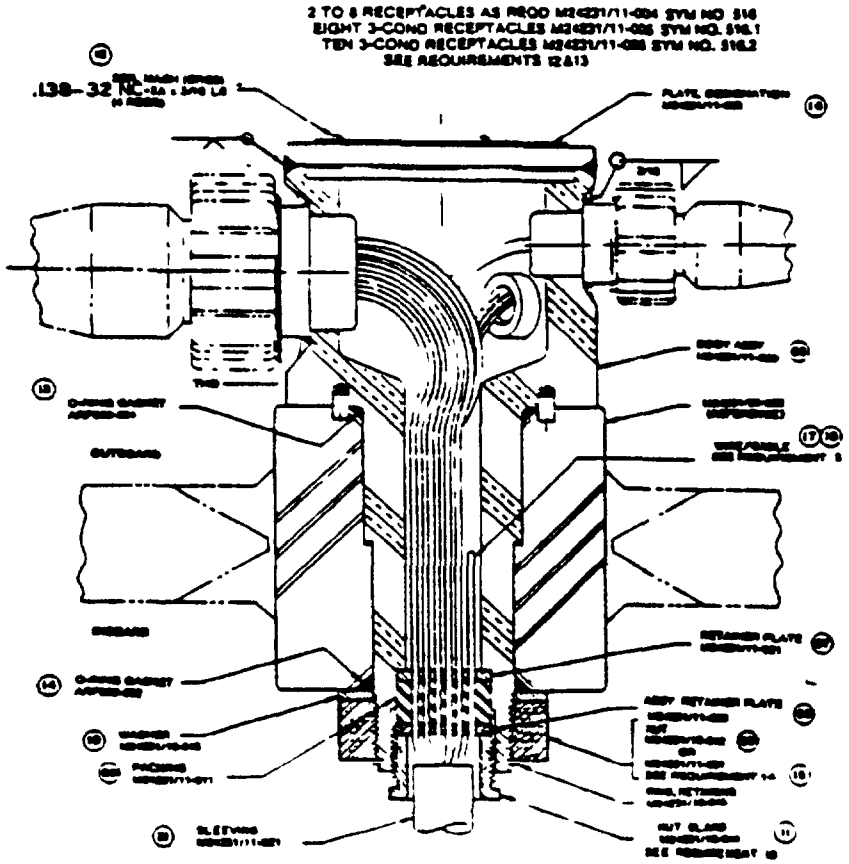
MIL-C-24231/11D(SH)
15 May 1987
SUPERSEDING
MIL-C-24231/11C(SHIPS)
30 August 1976
(See note 3)

MILITARY SPECIFICATION SHEET

CONNECTOR, MULTIPLE CABLE, TYPE IV

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the multiple cable connector described herein shall consist of this specification and the latest issue of MIL-C-24231.



SH 9678

FIGURE 1. Connector, multiple cable (two-piece body). (01) (02) (03)

AMSC N/A FSC 5935

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

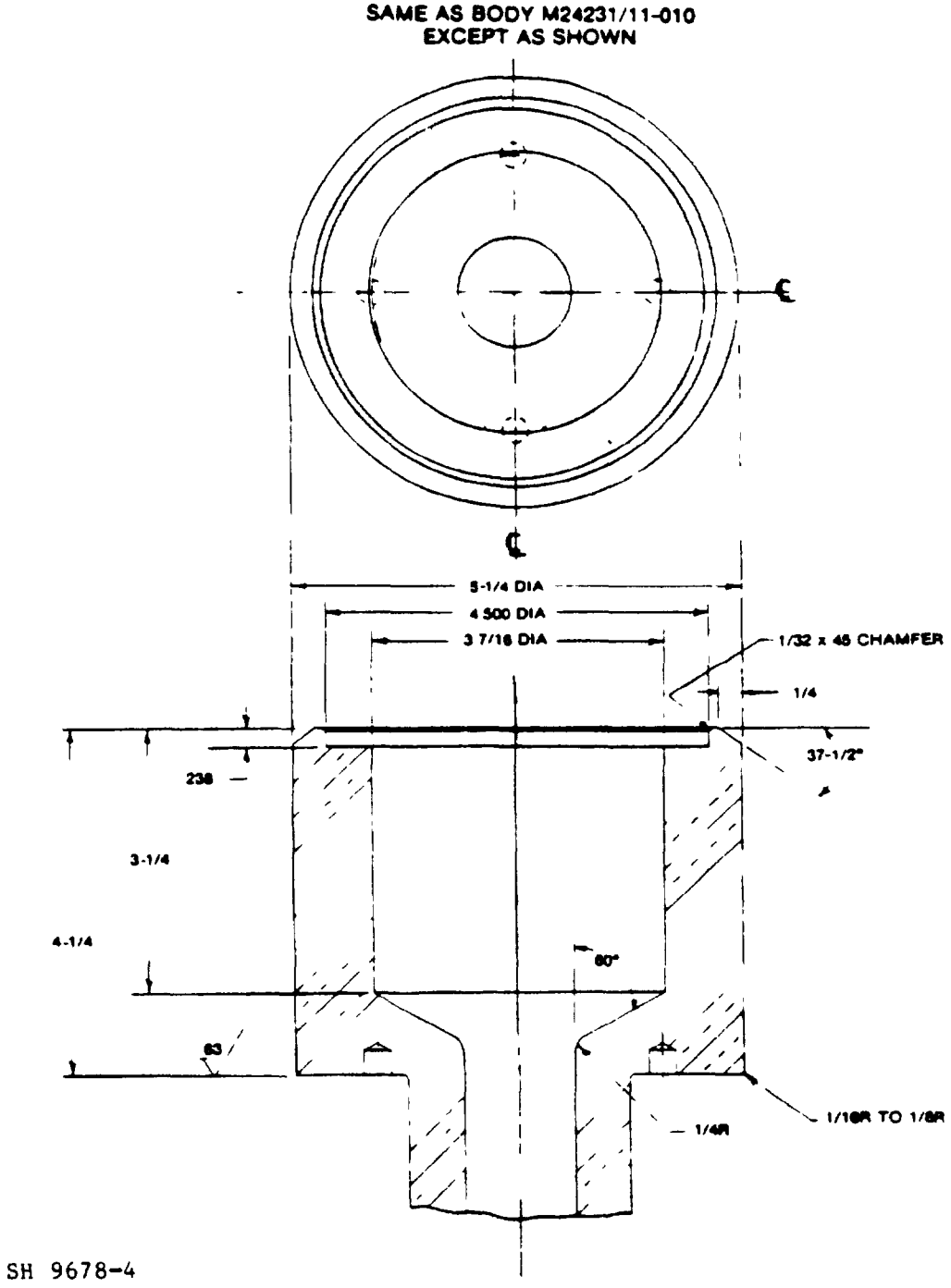
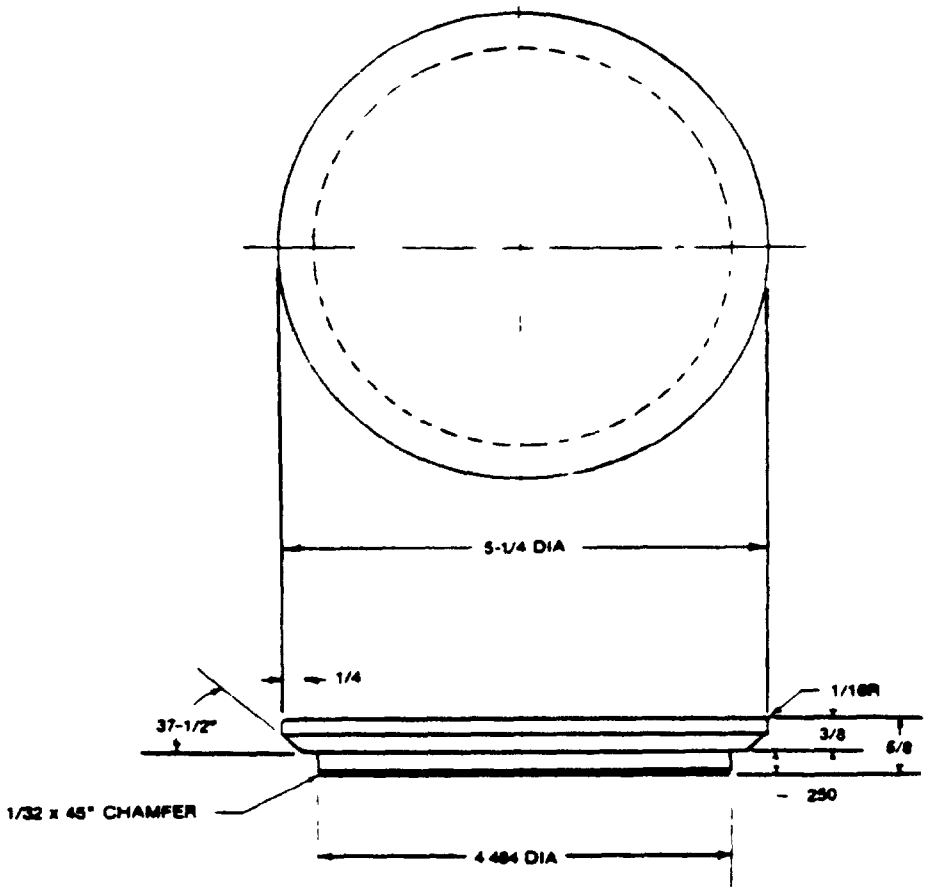


FIGURE 2. Body - M24231/11-007. (19)

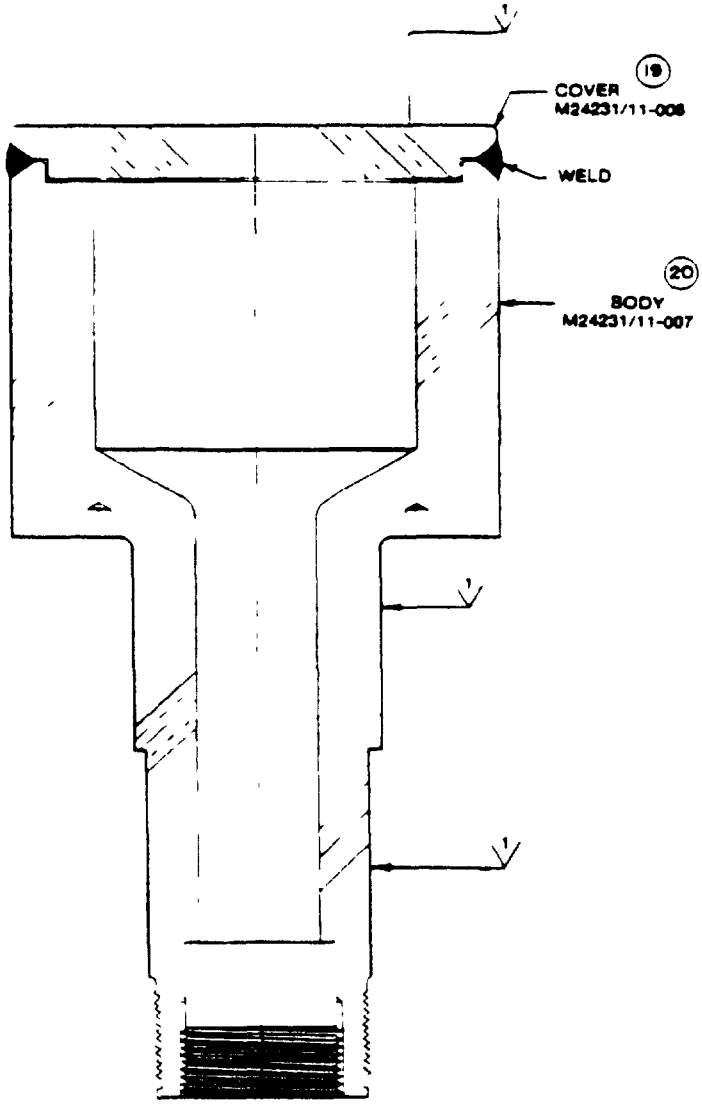
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



SH 9678-5

FIGURE 3. Cover M24231/11-008 for two-piece body. (20)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

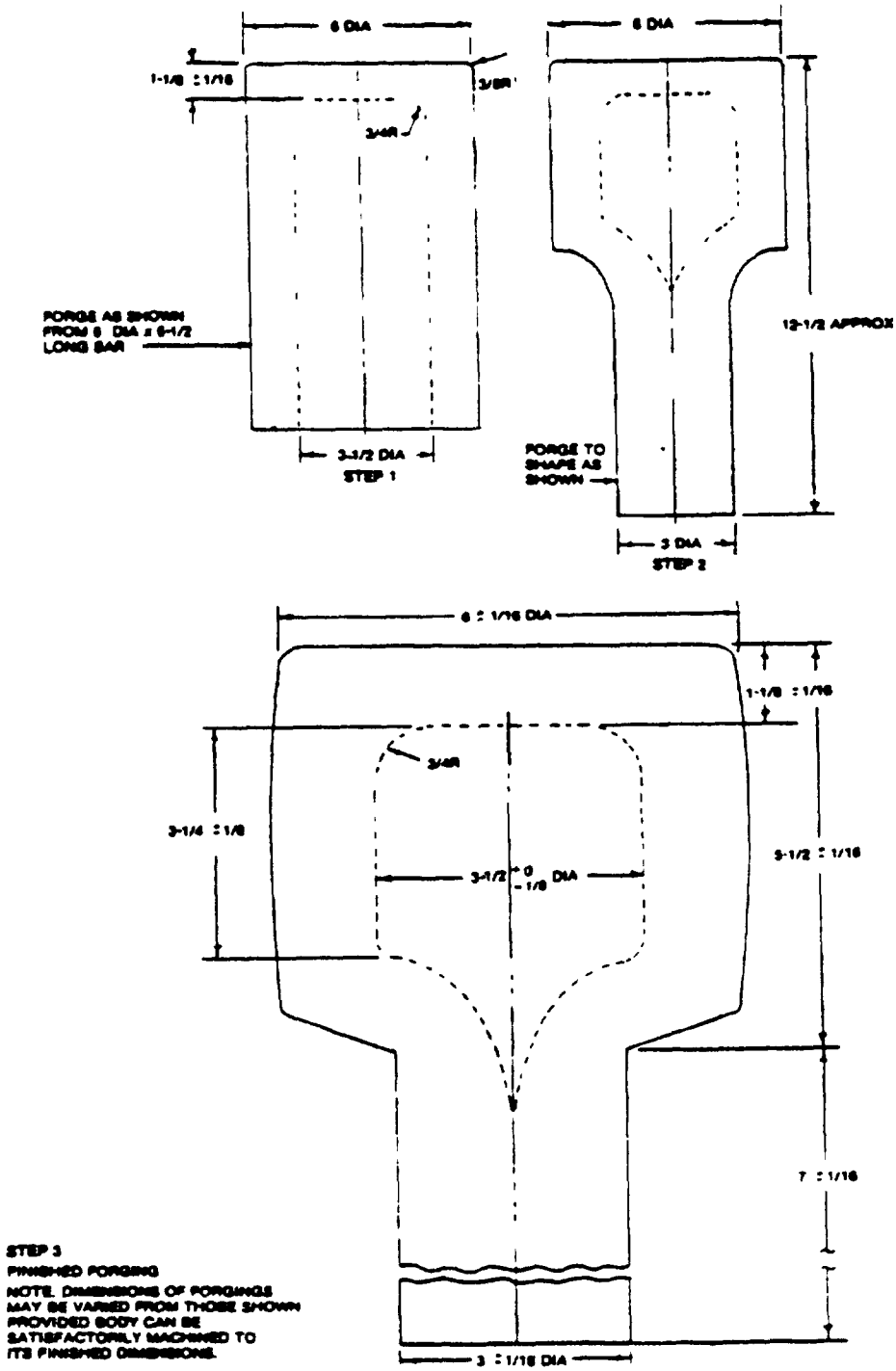


1/ HARDNESS TO BE TAKEN AT THESE POINTS

SH 9678-6

FIGURE 4. Body assembly - M24231/11-009. (05)

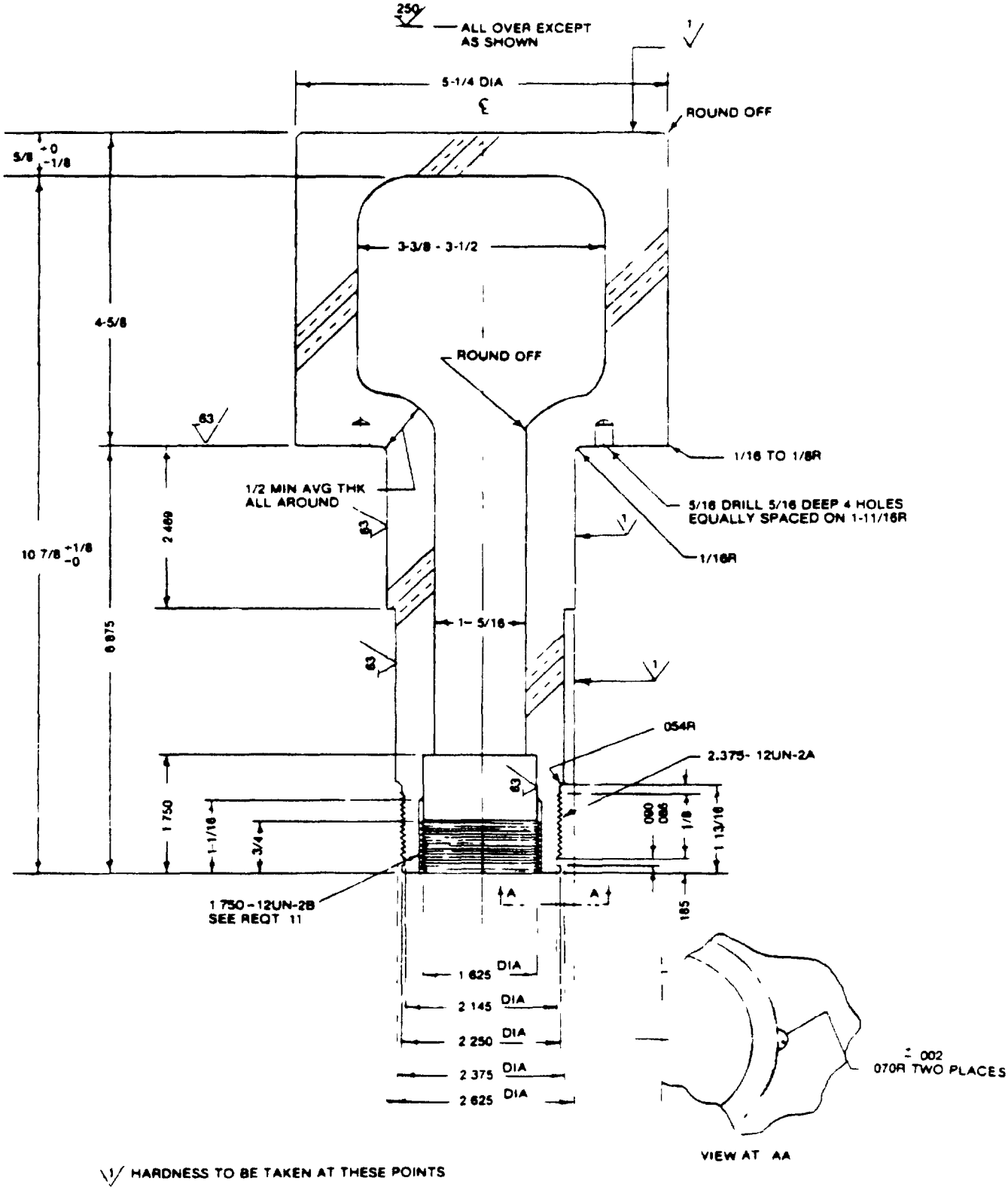
Source: https://assist.dla.mil -- Downloaded: 202610-19T20:14Z
Check the source to verify that this is the current version before use.



SH 9678-1

FIGURE 5. Forgings.

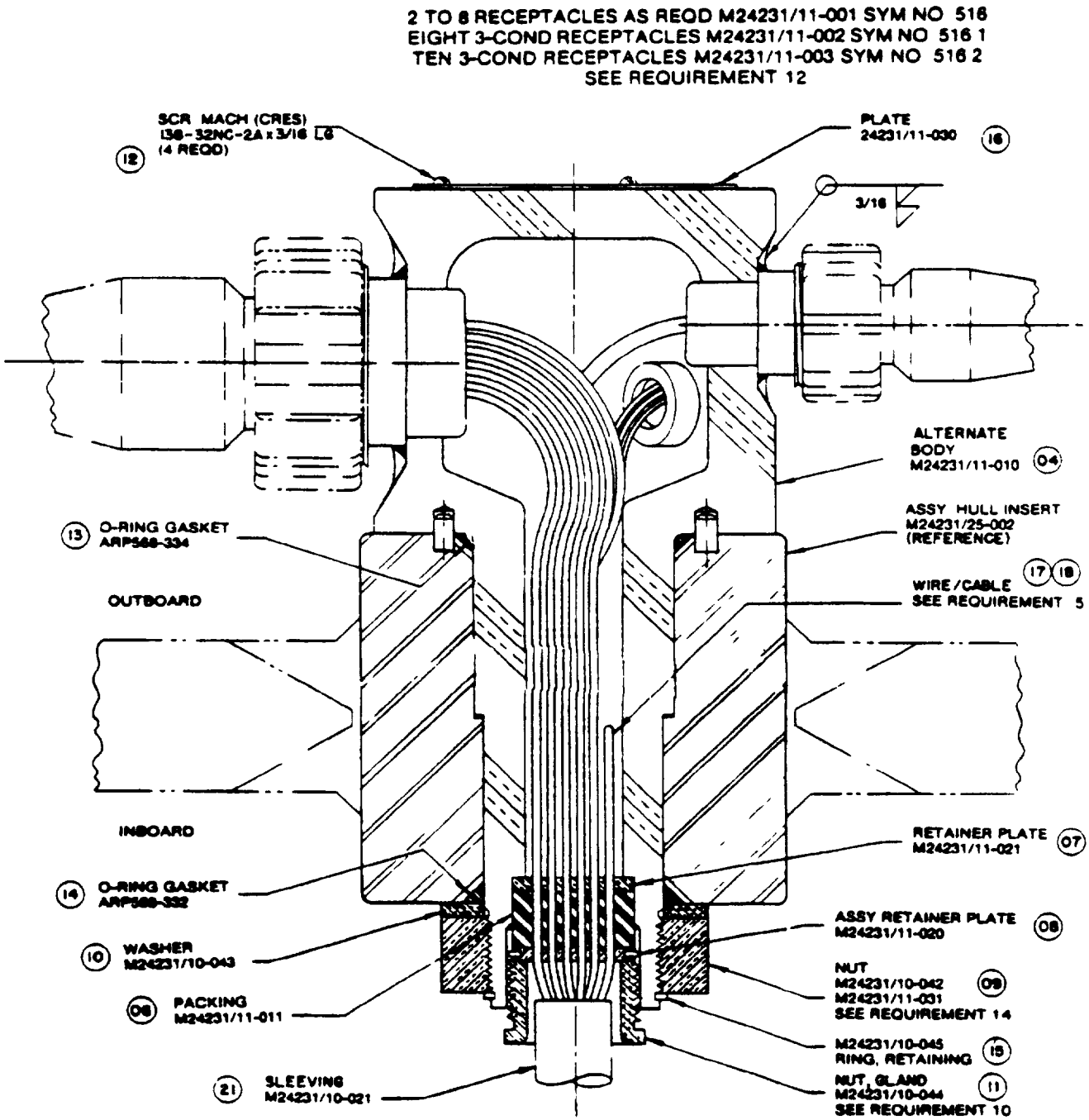
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



SH 9678-2A

FIGURE 6. Alternate body - M24231/11-010.

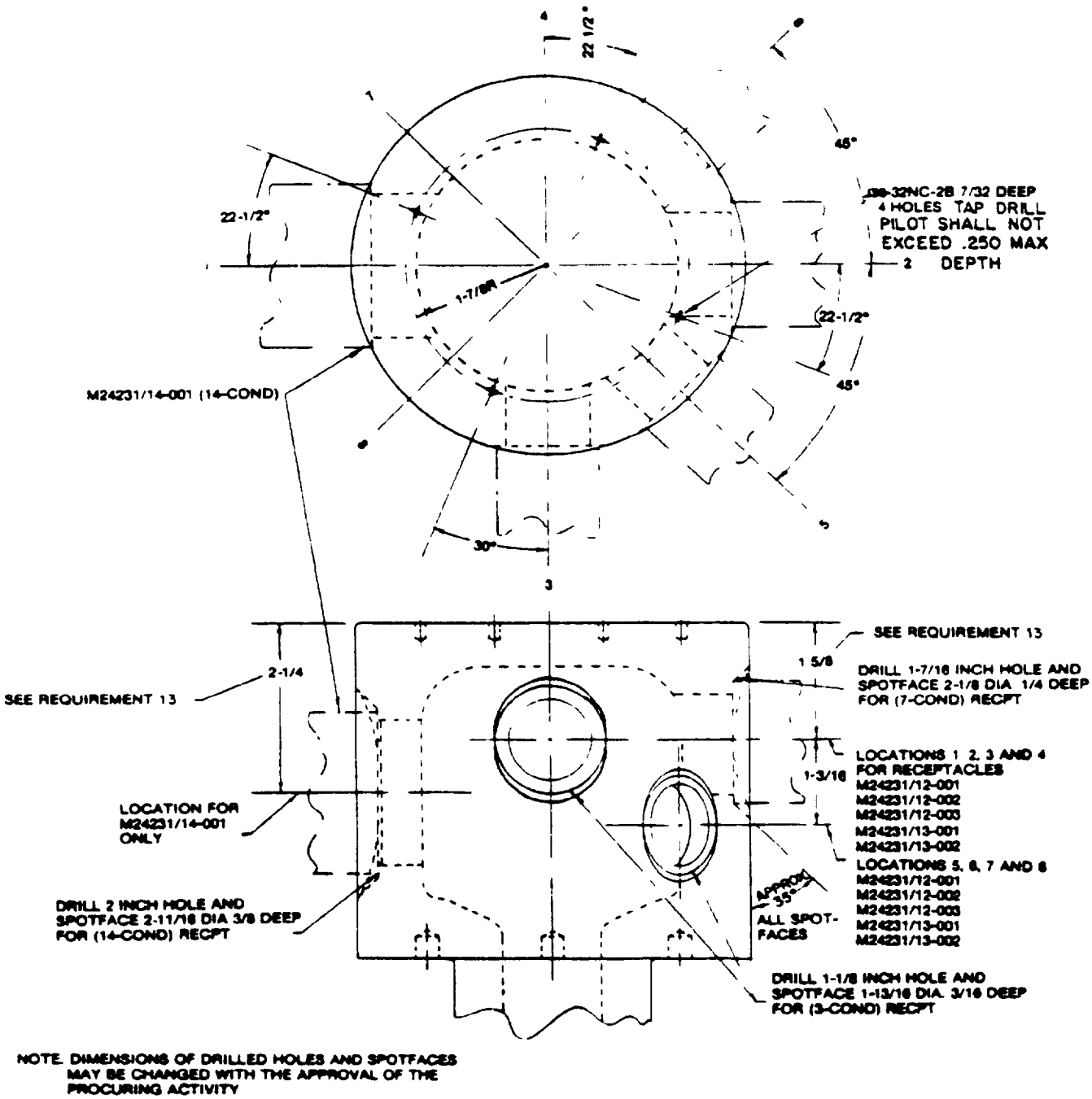
Source: https://assist.dla.mil -- Downloaded: 202610-19T20:14Z
Check the source to verify that this is the current version before use.



SH 9678A

FIGURE 7. Connector, multiple cable (alternate one-piece body). 01 02 03

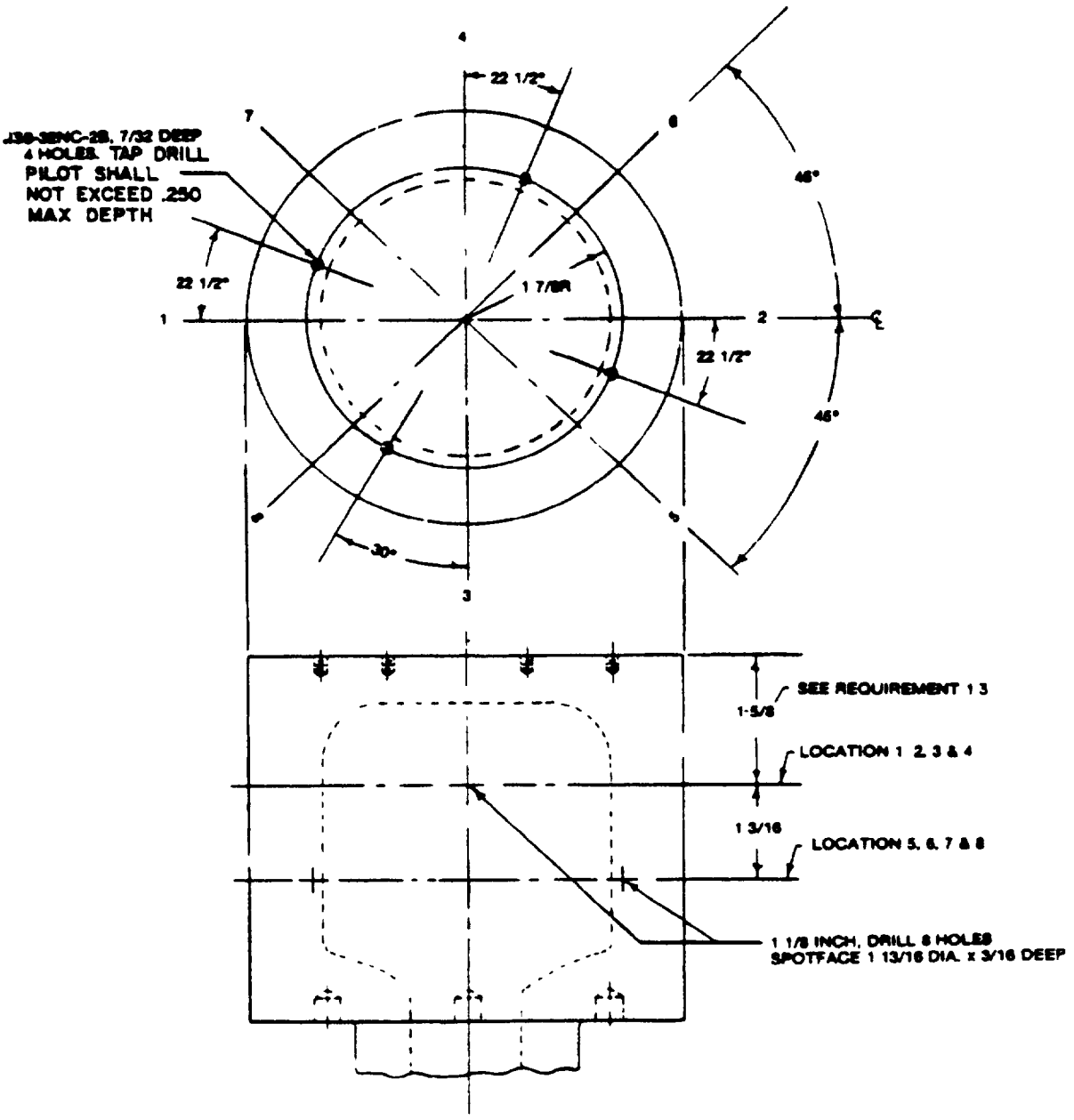
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



SH 9678-7

FIGURE 8. Typical location of receptacles and drilling sequence for M24231/11-001 and M24231/11-004 (see requirement 6).

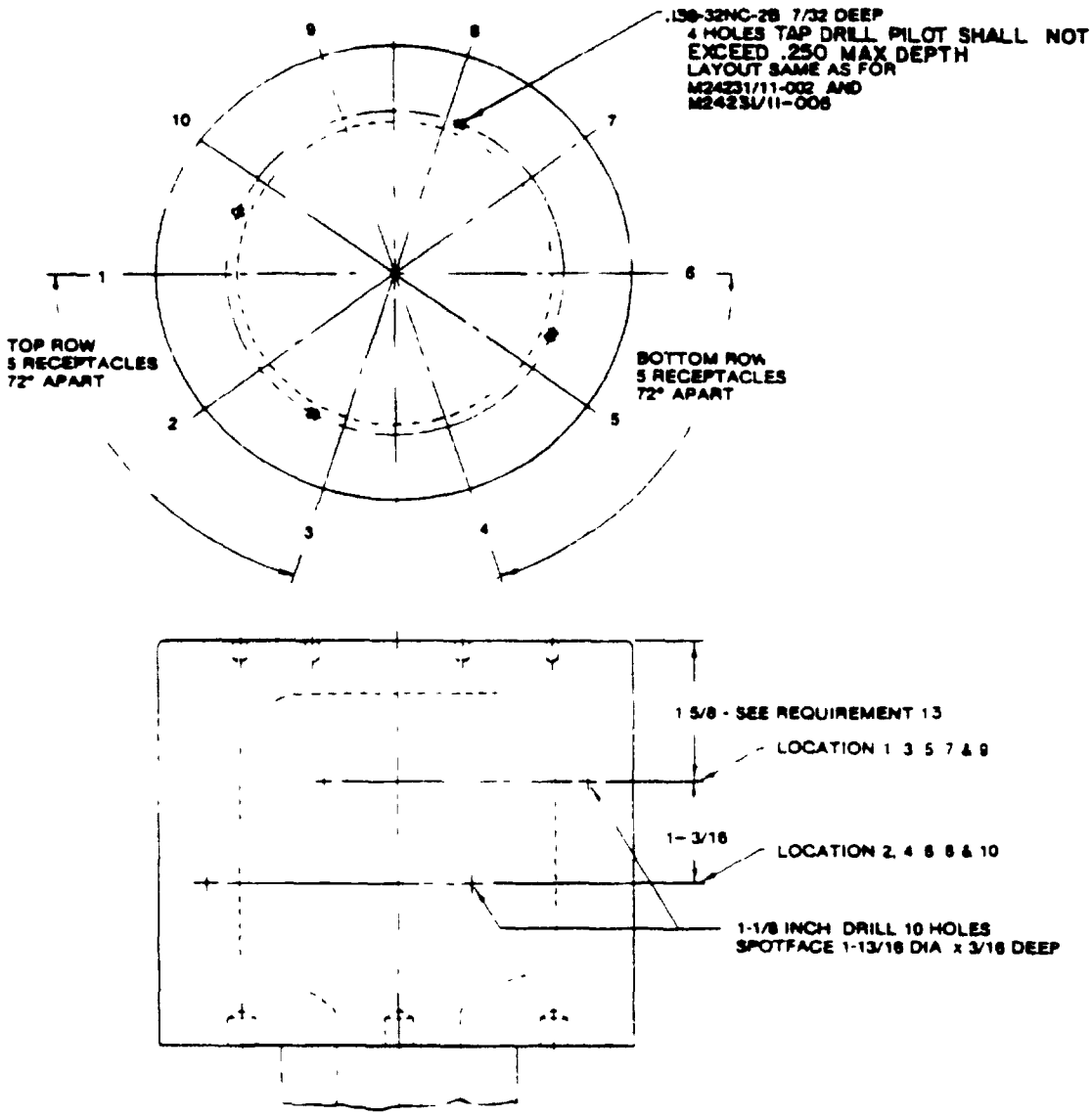
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



SH 9678-8

FIGURE 9. Location of receptacles for M24231/11-002 and M24231/11-005.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



SH 9678-9A

FIGURE 10. Location of receptacles for M24231/11-003 and M24231/11-006.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

M24231/11-021 SAME AS M24231/10-081 EXCEPT
HOLE SIZES AND LOCATIONS AS SHOWN
M24231/11-020 SAME AS M24231/10-080 EXCEPT
HOLE SIZES AND LOCATIONS AS SHOWN
M24231/11-011 SAME AS M24231/10-082 EXCEPT
HOLE SIZES AND LOCATIONS AS SHOWN

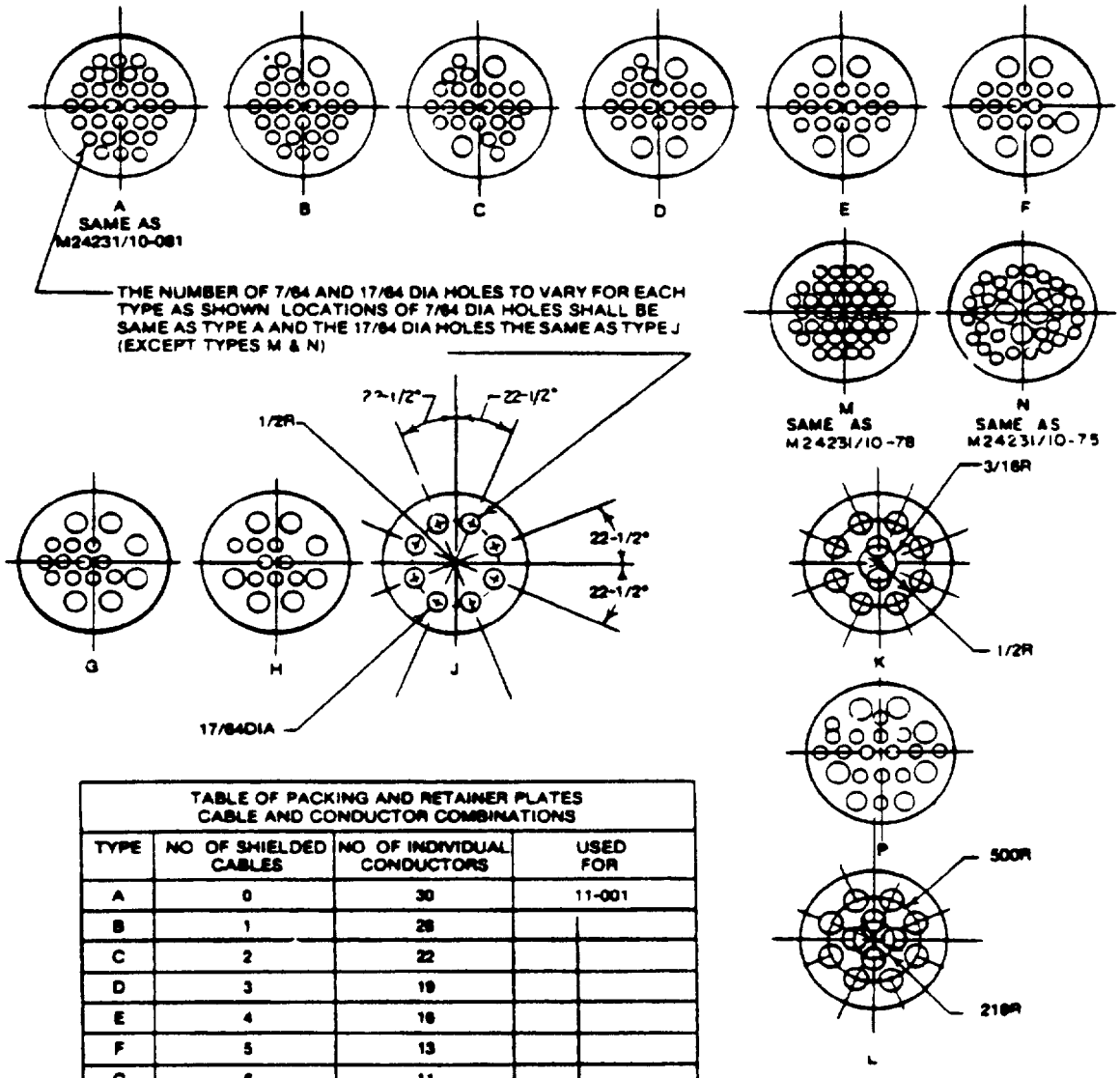
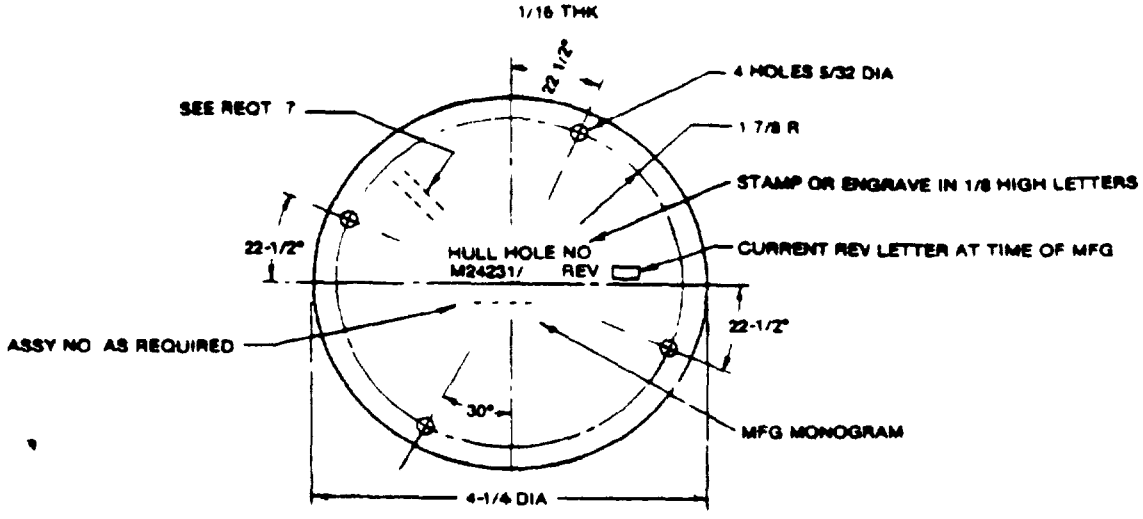


TABLE OF PACKING AND RETAINER PLATES CABLE AND CONDUCTOR COMBINATIONS			
TYPE	NO OF SHIELDED CABLES	NO OF INDIVIDUAL CONDUCTORS	USED FOR
A	0	30	11-001
B	1	26	
C	2	22	
D	3	19	
E	4	16	
F	5	13	
G	6	11	
H	7	8	
J	8	0	11-001 11-002
K	10	0	11-003
L	12	0	
M	0	40	
N	4	30	
P	7	15	11-001

SH 9678-10

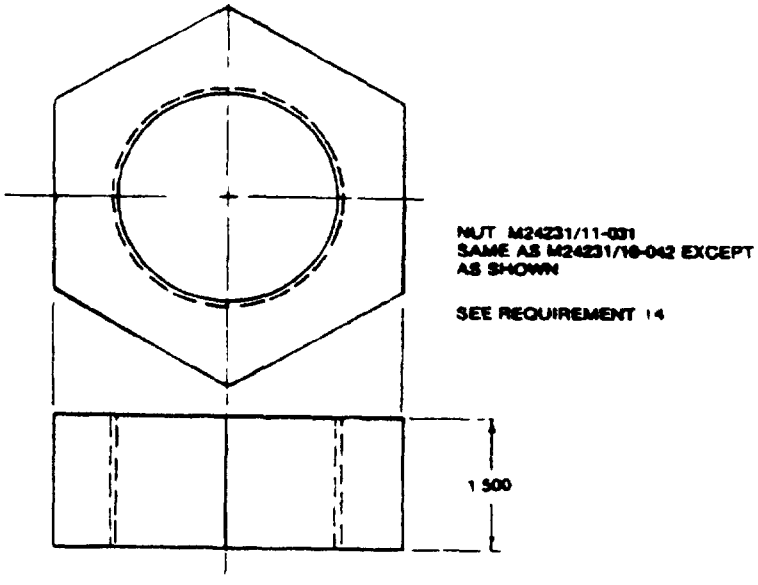
FIGURE 11. Retainer plates and packings. (06) (07)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



NOTE DIMENSION AND SPOTFACES MAY BE CHANGED WITH THE APPROVAL OF THE CONTRACTING ACTIVITY

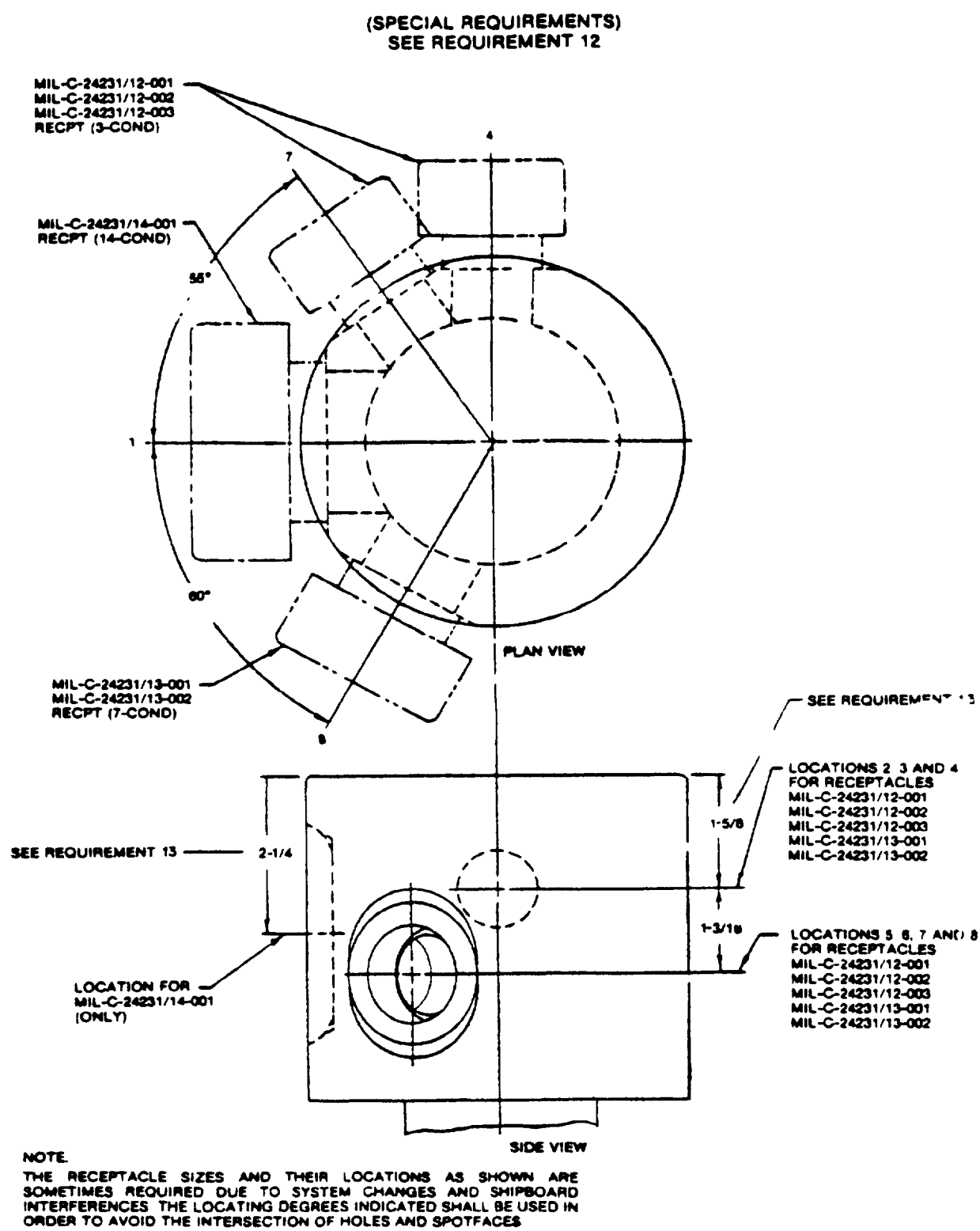
FIGURE 12. Designation plate - M24231/11-030. (16)



SH 9678-11

FIGURE 13. Nut - M24231/11-031. (09)

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:14Z
Check the source to verify that this is the current version before use.



SH 9678-12

FIGURE 14. Location of receptacles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Dimensions and configuration: See figures.
2. The body shall be manufactured from Ni-Cu-Al alloy in accordance with QQ-N-286, forging quality. The body shall be forged as shown or manufactured as a two-piece welded assembly. Exposed surfaces of the forging shall be visually examined. If cracks, seams, laps, burnt spots or any defect that will affect the performance of the finished body are discovered, it will be rejected. Machining shall be in accordance with requirement 3.
3. Recommended sequence of the machining shall be as follows:
   (a) Chuck 3-inch diameter tailpiece and machine head.
   (b) Chuck head and drill 1-5/16 hole. Locate end of tailpiece 10-3/8 inches from the end of the cavity as shown herein, and measure length dimensions from this point.
   (c) After completion of machining, but prior to drilling holes for receptacles, visual and dimensional examination and hardness and dye penetrant test for multiple cable connectors shall be performed.
4. Receptacles shall be installed in the body by the manual metal arc or tungsten inert gas welding methods and in the sequence specified in requirement 6, and as follows:
   (a) After completion of all machining (if accomplished in the annealed condition), the body shall be heat treated to obtain the physical properties listed in 3.3.2 (see requirements 15 and 16). Nickel-copper alloy (Ni-Cu) and Ni-Cu-Al alloy (K-monel) are subject to attack by sulphur at elevated temperatures. Material which might contain sulfur such as oil, grease, paint, marking pencil marks, lubricants, and other such material shall be completely removed prior to heat treatment. Atmospheres used for heating shall also be relatively sulfur free.
   (b) Applies to all other Ni-Cu and Ni-Cu-Al alloy parts.
   (c) Cleaning accomplished by the vapor de-grease method or equal.
   (d) Clean prior to welding.
   (e) Clean prior to heat-treatment (if required).
   (f) Size of fillet weld 3/16 inch. Total of 3 beads.
   (g) Liquid penetrant inspect (P.T.) root and final weld.
   (h) Weld applied in quadrants 3/4 inch (maximum) long opposite each other until weld is complete.
   (i) Caution shall be exercised during welding procedure to control temperature build-up in order to prevent any distortion of receptacles.
   (j) Recommended inter-pass temperature is 80 to 90°F.
5. Any unused holes in retainer plates and packing shall be plugged by wire, type B-16 with nylon jacket (see MIL-W-16878/1) for 7/64-inch diameter holes, and cable (see MIL-C-24643/33) for 17/64-inch diameter holes. Wires or cable shall be passed through two holes with loop on inside of fitting. Single holes shall be plugged with wire or cable with the end inside of fitting secured by a knot or other method approved by the contracting activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (Continued)

6. Two to 10 receptacles, as required, shall be installed in sequence in numbered locations shown herein. The larger receptacles shall be located in the lowest consecutive numbers.
7. Reserved for the installing activity to stamp or engrave circuit identification directly over the respective receptacle.
8. Wires shall be secured in groups with tape beyond the end of the sleeving, according to the receptacle, and marked with pin and circuit designation numbers. Shielded cable shall also be marked with pin circuit identification numbers at the exposed end.
9. Unless otherwise specified in the contract or order, the length of wires or cables extending from the end of the fitting shall be a minimum of 13 feet. Wires and cables shall be protected after installation by a method approved by the contracting activity.
10. The gland nut shall be tightened until the packing tightens against the wires and cable. This can be determined by moving the wires and cables gently back and forth until the wire and cable is seized by the packing. The gland nut should be turned approximately one-fourth of a turn more to hold wires and cables in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires and cables.
11. Full or partial threads shall not extend more than 0.083 inch beyond the 3/4-inch thread length.
12. Unless otherwise specified in the contract or order, all receptacles shall be positioned with the polarizing slot in the "DOWN" (6 o'clock) position plus or minus 2 degrees.
13. When the multiple cable connector is manufactured using the two-piece construction, all measurements for the location of the receptacles shall be made from the top of the cover for installation of M24231/13 or M24231/14 receptacles, the 1-5/8 inch and the 2-1/4 inch receptacle locating dimensions shall be increased to 1-7/8 and 2-3/8 inches respectively in order to avoid spot-face interference with cover weld.
14. Revision C of this specification modified the body, M24231/11-010 for assembly with nut M24231/11-031. Existing nut (REV B) M24231/10-042 is retained for use as a replacement part for existing bodies, in service, manufactured to prior revisions.
15. Welding of the cover (MIL-C-24231/11-008) and the receptacles shall be accomplished in accordance with either of the following:
    (a) While the body is annealed (prior to age-hardening).
    (b) After the body and cover have been age-hardened.
16. Hull fitting body-to-cover weld may be accomplished using either of the following welding electrodes or rods in accordance with MIL-E-21562:
    (a) MIL-EN-60 or MIL-RN-60.
    (b) MIL-EN-64 or MIL-RN-64.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

TABLE I. Parts list.

Item 1/ no.	Description 2/	Quantity for M24231/11- 3/						Material/specification	Remarks
		001	002	003	004	005	006		
01	Assembly, multiple cable, connector, symbol no. 516	1			1			-----	2-8 receptacles, see figures 1 and 7
02	Assembly, multiple cable, connector, symbol no. 516.1		1			1		-----	(8) - 3-conductor receptacles see figures 1 and 7
03	Assembly, multiple cable, connector, symbol no. 516.2			1			1	-----	(10) - 3-conductor receptacles see figures 1 and 7
04	Body, alternate	1	1	1				Ni-Cu-Al, QQ-N-286	See figure 8
05	Assembly, two-piece body		.		1	1	1	See items 19 and 20	See figures 2, 3, and 4
06	Class A, packing durometer 55 ± 5	1	1	1	1	1	1	Rubber, ZZ-R-768	See figure 11, type as required
07	Plate, retainer	1	1	1	1	1	1	Ni-Cu, class A, QQ-N-281	See figure 11, type as required
08	Assembly, retainer plate	1	1	1	1	1	1	Ni-Cu, class A, QQ-N-281	See figures 1, 7, and 11
09	Nut	1	1	1	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 13 and requirement 14
10	Washer	1	1	1	1	1	1	M24231/10-043	See figure 1
11	Nut, gland	1	1	1	1	1	1	M24231/10-044	See figure 1

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	Quantity for M24231/11- 3/						Material/specification	Remarks
		001	002	003	004	005	006		
12	Screw, machine, 0.138-32NC-2A by 3/16 long	4	4	4	4	4	4	CRES, com1	See figure 1
13	Gasket, O-ring	1	1	1	1	1	1	MS28775-334	See figure 1
14	Gasket, O-ring	1	1	1	1	1	1	MS28775-332	See figure 1
15	Ring, retaining	1	1	1	1	1	1	MS24231/10-045	See figure 1
16	Plate, designation	1	1	1	1	1	1	CRES or Al, MIL-P-15024	See figure 12
17	Wire	As required						Type B-16, with nylon jacket, MIL-W-16878/1	See figure 1 and requirement 5
18	Cable, 2SWU	As required						MIL-C-24643/33	See figure 1 and requirement 5
19	Body				1	1	1	Ni-Cu-Al, QQ-N-286	Part of assembly, two-piece body, see item 05
20	Cover, body				1	1	1	Ni-Cu-Al, QQ-N-286	Part of assembly, two-piece body, see item 05
21	Sleeving, 0.890 id, 0.050 thick, 10 ft long	1	1	1	1	1	1	Type F, form U, category 1, color clear, MIL-I-631	See figure 1

1/ Item numbers are the same as encircled numbers shown on figures.
2/ See applicable figure or table listed in remarks column for part numbers and dimensions.
3/ Quantity is for one complete assembly of each size.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

NOTES:

1. All dimensions are in inches.
2. Unless otherwise indicated, tolerances shall be as follows:

   (a) Decimals - plus or minus 0.005 inch.
   (b) Fractions - plus or minus 1/64 inch.
   (c) Angular dimensions - plus or minus 1/2 degree.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5935-N279-11)

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:14Z
Check the source to verify that this is the current version before use.